SORAIDA QUINTERO

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SUMMARY

Ph.D. Candidate with strong expertise in statistical analysis, data modeling, and interpreting complex datasets to support evidence based decision making. Proficient in R and experienced with Python and SQL for data extraction, transformation, visualization, and reporting. Skilled in identifying trends, developing analytical models, and translating quantitative findings into clear insights for academic leadership and non-technical audiences. Proven record of independent research, cross-unit collaboration, and managing data driven projects, with a commitment to improving student success and institutional outcomes through high quality analytics. Real Estate Managing Broker license with more than 15 years in the industry. Facilitated remote learning for students via Zoom and Microsoft Teams, utilizing breakout rooms and interactive polling to maintain high engagement levels.

EDUCATION

Ph.D. Candidate, University of Illinois Chicago *Fall 2026*
M.S., Purdue University *Dec 2021*
B.S., Chicago State University *May 2016*

Relevant Coursework:

Biostatistics, Statistical Methods Biology, Quantitative Methods, R programming, Stats for Industry, Stats for Natural Resources, SQL, Machine Learning, Agent & Individual-based modeling

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TECHNICAL SKILLS

Programming & Data:
R (tidyverse, dplyr, purrr, ggplot2, vegan); Python (pytorch, pandas, numpy, NLP/text extraction workflows); SQL, Bash scripting, Linux/Unix environments, GitHub, Jupyter Notebook
Statistical Modeling & Methods:
Generalized linear models (logistic & Poisson regression); mixed effects model interpretation, clustering, agent-based modeling, spatial statistics, simulation modeling (Monte Carlo, parallel), non-parametric analysis

Machine Learning & Predictive Analytics:
Supervised and unsupervised learning; regression and classification models; XGBoost gradient boosting, decision trees, random forest; model comparison; cross-validation; topic modeling; co-occurrence network analysis; NLP pipelines; image identification
Data Processing & Analysis:
Data wrangling, reproducible workflows, exploratory data analysis, text mining, structured and unstructured data integration, cleaning, validation, preprocessing, Excel, Word, PowerPoint
Languages:
Spanish (fluent)

RESEARCH EXPERIENCE

Graduate Research Assistant — University of Illinois, Chicago
2021 – Present

- Designed and validated agent-based and statistical models to evaluate how individual behaviors scale up to city wide level outcomes.
- Built generalized linear and logistic regression models to quantify how environmental and social factors influence spatial patterns.
- Conducted spatial statistical analysis (Moran's I, Mantel r) to identify clustering and decision mimicry.
- Performed regression and mixed-model analysis to evaluate socio-ecological responses to environmental change.
- Communicated analytical insights to interdisciplinary researchers, community stakeholders, and scientific audiences.

Graduate Fellow — Purdue University, Policy Research Institute, West Lafayette
2018 – 2021

- Managed project coordination and day-to-day operations for an interdisciplinary Mellon Foundation grant, including meeting facilitation, milestone tracking, and communication across faculty and administrative units.
- Developed stakeholder outreach plans and communication strategies to translate research findings into accessible formats for university leadership, policymakers, and external partners.
- Prepared formal reports, data summaries, and budget documentation to support institutional accountability and ensure compliance with foundation reporting requirements.
- Planned and coordinated events, workshops, and presentations to support interdisciplinary collaboration and increase visibility of research activities across campus.
- Collaborated with policy experts, communications teams, and Purdue University Press to produce high-quality research outputs and support evidence-based decision-making within the institution.

TEACHING EXPERIENCE (more upon request)

Teaching Assistant: Biostatistics with R — University of Illinois, Chicago
2021 – Present

- Supported undergraduate and graduate students in applying statistical methods including linear and logistic regression, ANOVA, non-parametric tests, and model diagnostics using R.
- Led hands-on instruction in data wrangling, exploratory data analysis, and statistical visualization to help students work with real-world datasets.
- Guided students in translating research questions into appropriate statistical models and interpreting results in a meaningful, applied context.
- Assisted with the design, grading, and feedback of coding assignments and projects, emphasizing reproducibility, clarity, and correct model interpretation.
- Mentored students in experimental design and quantitative reasoning, providing individualized support during office hours and project development.
- Helped students build confidence in statistical thinking and communicate analytical findings clearly to both technical and non-technical audiences.

SELECTED PROJECTS

Neighborhood Gentrification Predictions through Real Estate Market Analysis
Created a property analysis model to predict neighborhood gentrification patterns using property and area characteristics with publicly available housing and census data.
Methods: Regression and clustering in R using tidyverse, ggplot2, and machine learning algorithms (random forest, gradient boosting).
Outcome: (preliminary) Improved property value prediction accuracy and identified socio-economic indicators driving neighborhood change.

Spatial Mimicry in Chicago Residential Landscapes
Examined how social interactions and neighborhood factors shape spatial patterns of residential yard management.
Methods: Applied spatial statistics and logistic regression to assess plant clustering patterns by measuring spatial autocorrelation using Moran's I and Mantel r tests.
Outcome: Demonstrated that social behaviors at the neighborhood level are spatially autocorrelated, revealing mimicry patterns among adjacent households.

Free-Roaming Dog Population Modeling

Developed agent-based models to simulate dog population dynamics, and feeding behaviors across diverse ecological and cultural contexts.
Methods: Implemented demographic and movement modeling, intervention simulations (vaccination, sterilization, cultural), and spatial analyses
Outcome: (preliminary) Identified context-specific drivers of population level change, and demonstrated that targeted interventions for free-roaming dogs yield significantly improved management outcomes.

PUBLICATIONS

Quintero S, Ballut N, Johnson P, Minor E, (2026). *Examining the mechanisms that create spatial patterns in vegetation across residential landscapes.* Computers Environment and Urban Systems. Manuscript submitted.

Quintero S, Zollner P, Atallah S, (2026). *Understanding the Disturbance of Human Recreation on Wildlife using Multiple Dynamic Agents within an IBM Framework.* Ecological Modelling. Manuscript in progress.

Quintero S, Weldon L, Clawson R, Roberts A, (2026). *Engaging Communication for High Impact Research: Four models for organizing interdisciplinary research teams*. Proceedings of the National Academy of Sciences. Manuscript in progress.

Quintero S, Minor E, Railsback S, Johnson P, (2023). *Yards* (Version 1.0.0). CoMSES Computational Model Library.

Quintero S, (2022). *Making the change.* The Wildlife Professional, 16(2), 49–51.

Hernandez G, Quintero S, Vilela J. F., & de La Sancha N. U, (2017). *Ontogenetic variation of an omnivorous generalist rodent: The case of the montane akodont* (*Akodon montensis*). Journal of Mammalogy.

CONFERENCE PRESENTATIONS & POSTERS

Quintero S, Minor E. Neighbor Mimicry and Plant Nurseries as Drivers of Spatial Patterns in Urban Residential Landscapes. IALE Conference; 2023 March 19; Riverside, California.

Quintero S, Cohen A , Atallah S, Zollner P. Understanding the Disturbance of Human Recreation on Wildlife Using Multiple Dynamic Agents Within an IBM Framework. BTAA GIS Conference; 2020 Nov 13; Virtual

Quintero S, Cohen A , Atallah S, Zollner P. Understanding the Disturbance of Human Recreation on Wildlife Using Multiple Dynamic Agents Within an IBM Framework. Poster presentation. The Wildlife Society Conference; 2020 Sept 28 – Oct 2; Virtual

Quintero S, Weldon L, Clawson R, Roberts A. Engagement models of science communication: *Working tools for team researchers at a micro level.* Poster presentation at: International Network for the Science of Team Science. SciTS Conference; 2019 May 20-23; Lansing, MI. ***Best poster award***

Quintero S, Kirt S. Ant's in the Prairie: A comparison of Midewin National Tallgrass Prairie's Remnant and Restored Areas. Poster presentation at: LSAMP International Undergraduate Research Conference, 2014; Tarlac, Philippines

Quintero S, Kirt S. Ant's in the Prairie: A comparison of Midewin National Tallgrass Prairie's Remnant and Restored Areas. Oral presentation at: Argonne National Laboratory, 2013; Darien, IL.

WORKSHOPS, FELLOWSHIPS & AWARDS

Andrew Mellon Fellow • Alliance for Graduate Education Scholar • ESA Seeds Grant • NCEAS Director's Scholar • DaSEH, R Workshop, Fred Hutch Cancer Center

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PROFESSIONAL MEMBERSHIPS

Ecological Society of America • American Society of Mammalogists • Pycon • UIC Biology Graduate Student Association • Chicago Association of Realtors

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OUTREACH & ENGAGEMENT

UIC Diversity Equity Justice & Inclusion • Mujeres Latinas en Acción (Battered Women's Shelter) • Great Lakes Coastal Restoration • Prairie Conservation & Community Education

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REFERENCES

Dr. Emily Minor
Professor, University of Illinois Chicago
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Dr. Pat Zollner
Professor, Purdue University
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Dr. Gaby Nunez-Mir
Assistant Professor, University of Illinois Chicago
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